Fiery DataXPress Sample FITS-Client

FITS DataPath Example





Revision History

Version	Change Log	Date	Author
1.0	Initial Draft	04/23/2024	Scott Crosby

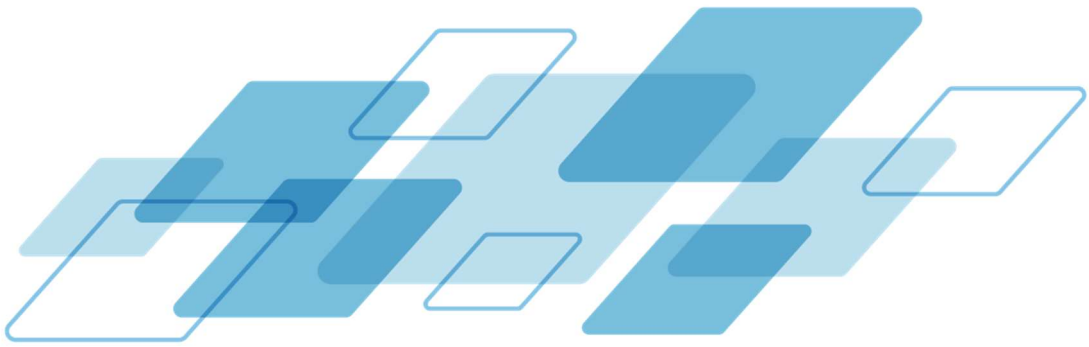




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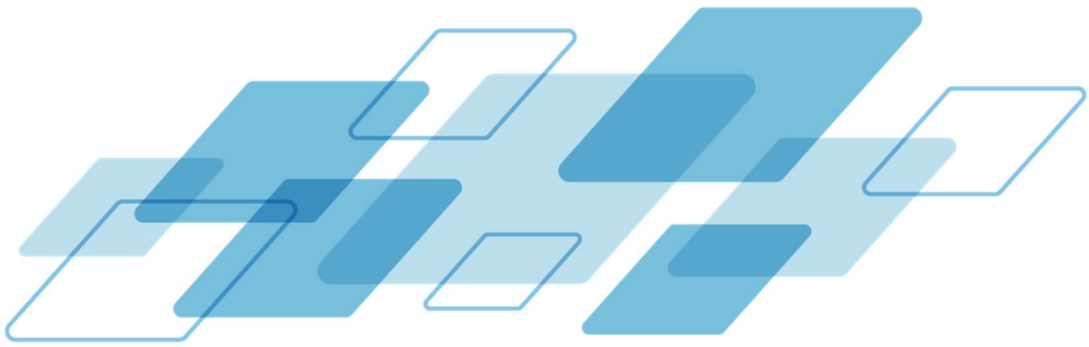
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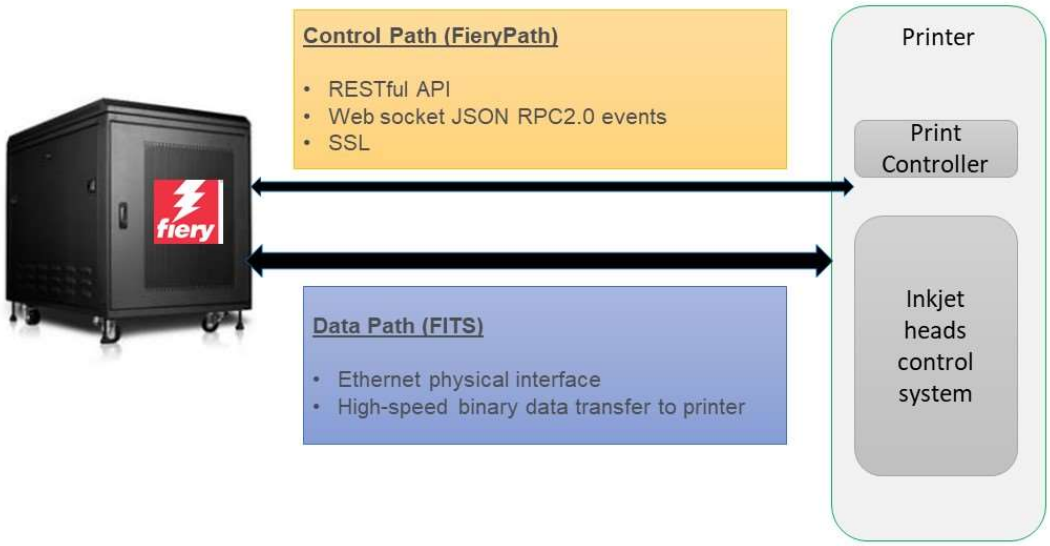


1 Overview

This document is intended to serve as a reference for a printer to interface with the Fiery DFE's Data Path (using the FieryPath Image Transfer Service specification). Fiery DataXPress (FDX) provides a framework that printer clients can leverage – either to build upon the sample clients provided or use them as a reference to create their own clients. The goal of FDX is to demonstrate how the printer can create clients to print at speed.

1.1 Data Path

The Data Path is the interface for handling the transfer of image data from the DFE to the printer. The Control Path, by receiving events and acknowledging them using the documented RESTful API, allows data to enter the Data Path. The FieryPath Image Transfer Service (FITS) specification gives more information about FITS and the protocol used for data transfer from the DFE to the printer.



2 Design

The sample FDX FITS-Client is broken up into three components: FITS-Core, FITS-Module, and FITS-Output. The sample client implements the FITS protocol -- connecting to a Fiery FITS-Server and receiving image data from the Fiery. The client also implements buffering and communicates with the sample FDX Control Path client to manage “print” and “cancel” requests. In addition, the sample FITS-Client provides several examples that demonstrate “processing” the image data (prior to being stored in memory) and “transferring” the image data (i.e. writing the image data to files).

2.1 FITS-Core

The FITS-Core implements the FITS protocol as described in **FieryPath_FITS** specification. It is responsible for managing the connection to a Fiery FITS-Server. That includes connecting to the Fiery FITS-Server, sending the CLIENT_CONNECTED message, reading the SERVER_CONFIG message, and connecting to the supported “colorant” sockets. Once connected, the FITS-Core will receive image data on the various “colorant” connections, handing them off to the FITS-Module for further processing. The FITS-Core also implements the logging component that is used not only by the various FITS-Core components, but also the FITS-Module and FITS-Output plug-ins.

At start-up, the FITS-Core initializes the logging component and then attempts to load the FITS-Module. If successful, it will initialize the FITS-Module and start-up the Fiery FITS-Server connection and FITS-Module. The FITS-Core will then run continuously, receiving image data from the Fiery server and handing it off to the FITS-Module, until the FITS-Client exits/quits.

2.2 FITS-Module

The FITS-Module implements image data management. It is responsible for storing/buffering image data received from the FITS-Core, managing the connection to the FDX Control Path client (including reading “print” and “cancel” messages, and sending “printed” responses), and “outputting” the image data to the currently loaded FITS-Output plug-in.

At start-up the FITS-Core loads and initializes the FITS-Module. Once the FITS-Core starts-up the Fiery FITS-Server connection, it will then start-up the FITS-Module for all colorants connected to the FITS-Core. This includes “loading” the appropriate FITS-Output plug-in. If successful, the FITS-Module will then allocate a memory pool (per colorant) for storing the image data and starting-up the connection to the FDX Control Path client. As image data is received by the FITS-Core and handed off to the FITS-Module, it will check whether the loaded FITS-Output plug-in wants to “process” the image data before being stored in the allocated memory pool. As the memory pool fills up, the FITS-Module will “throttle” the FITS-Core by blocking until enough memory is available to store the next image. Meanwhile, as the FDX Control Path client receives “imageCompleting” FieryPath events, it will send corresponding “print” messages to all connected FITS-Modules – causing FITS-Module to hand off the stored (and potentially “processed”) image data to the FITS-Output plug-in for “transfer”. This “transfer” comes in two flavors – either each colorant’s image data for a given page is “transferred” independently (each colorant on its own thread) or the image data for the given page can be “accumulated” and “transferred” altogether (in a single thread). Once the “transfer” is complete, the image data is removed from the memory pool – freeing up space for more image data. If the FDX Control Path client sends a “cancel” message, all image data associated with the job being canceled is removed from the memory pool (without being “transferred”).

2.3 FITS-Output

The FITS-Output implements the specific image data requirements for a given press/printer. At start-up, the FITS-Module attempts to load the FITS-Output plug-in. If successful, it will initialize the FITS-Output plug-in according to the configuration settings for the FITS-Output plug-in.

Once initialized/configured, each plug-in can support one or more of the following FITS-Output features:

- Process: Allows a FITS-Output plug-in to “process” the given image data before being stored in memory.
- Transfer: Allows a FITS-Output plug-in to “transfer” the given image data for a given colorant.
- Transfer (Accumulated): Allows a FITS-Output plug-in to “transfer” the given image data for all colorants in a single thread.
- Wait for Ejected: Allows a FITS-Output plug-in to “wait” for the given image to be “ejected” by the electronics of a press/printer – ensuring the image will not need to be recovered/reprinted if a problem arises on the printer.

2.3.1 FITS-Output-Raw

FITS-Output-Raw is the simplest of the pre-built FITS-Output plug-ins provided. The image data is not “processed” before being stored in memory, but it can be configured to use either the “transfer” or the “transfer (accumulated)” interface. If enabled (via the configuration/settings described in the next section), this plug-in can write the image data for each page’s colorant to a file.

2.3.2 FITS-Output-Tiff

FITS-Output-Tiff supports the generation of TIFF image data. The image data for each colorant is “processed”, generating the corresponding TIFF data, before being stored in memory. This plug-in can be configured to use either the “transfer” or the “transfer (accumulated)” interface. If enabled (via the configuration/settings described in the next section), this plug-in can write the image data for each page’s colorant to a file.

2.3.3 FITS-Output-Prn

FITS-Output-Prn supports the generation of PRN page data. The image data is not “processed” before being stored in memory and it only uses the “transfer (accumulated)” interface – because all colorants are required to generate PRN page data. If enabled (via the configuration/settings described in the next section), this plug-in can write each page’s data to a file.

2.3.4 FITS-Output-Simulator

FITS-Output-Simulator supports simulating the delays inherent with transferring image data to the electronics of a press/printer. The image data is not “processed” before being stored in memory, but it can be configured to use either the “transfer” or the “transfer (accumulated)” interface. If enabled (via the configuration/settings described in the next section), this plug-in can simulate a delay (in milliseconds) for each “transfer” or “transfer (accumulated)” invoked. In addition, if enabled, this plug-in can simulate a page rate (in images/minute) for each “wait for ejected” invoked.

3 Configuration

The configuration of the sample FDX FITS-Client is managed through a JSON file – FITS-Module.json. Each component has settings that are controlled by this file.

3.1 FITS-Core

The following table lists the JSON elements used to configure the FITS-Core.

Element	Type	Required	Purpose
fits_server	JSON Object	Yes	JSON Object containing the attributes related to the FITS-Server.
ip	String	Yes	Indicates the IP of the FITS-Server to connect to.
port	Number	No	Indicates the port of the FITS-Server to connect to. Default value is **9210**.
logging	JSON Object	No	JSON Object containing the attributes related to logging in the FITS-Client.
console	Boolean	No	Indicates whether certain critical logs should also be displayed on the FITS-Client console. Default value is **false**.
max_files	Number	No	Indicates the maximum number of log files created before log "rotation" kicks-in. Default value is **20**.
max_size_in_megabytes	Number	No	Indicates the maximum log file size (in MB). Default value is **20**.

Table 1: FITS-Core Settings

3.2 FITS-Module

The following table lists the JSON elements used to configure the FITS-Module.

Element	Type	Required	Purpose
colorant	JSON Object	Yes	JSON Object containing the attributes related to the supported colorants.
buffer_pool_size_in_megabytes	Number	Yes	Indicates the IP of the FITS-Server to connect to.
fiery_path_client	JSON Object	Yes	JSON Object containing the attributes related to the FieryPath Client.
ip	String	Yes	Indicates the IP of the FieryPath Client to connect to.
port	Number	No	Indicates the port of the FieryPath Client to connect to.
output	JSON Object	Yes	JSON Object containing the attributes related to the output module.
plugin	String	Yes	Indicates the name of the FITS-Output plug-in to load.

Table 2: FITS-Module Settings

3.3 FITS-Output

The sample FDX FITS-Client supports several pre-built FITS-Output plug-ins. The following tables list the JSON elements used to configure the various FITS-Output plug-ins.

Element	Type	Required	Purpose
output_raw	JSON Object	No	JSON Object containing the attributes related to the FITS-Output-Raw plug-in.
directory	String	No	Indicates the directory where image data files are written (if enabled). Default value is "**./raster**".
file_enabled	Boolean	No	Indicates whether the image data is written to file. Default value is **false**.
job_name_enabled	Boolean	No	If writing to file, indicates whether the Fiery job name is used in the name of the output file. Default value is **false**.
colorant_accumulator_enabled	Boolean	No	Indicates whether the image data is "accumulated" and processed in a single thread. Default value is **false**.
wait_for_ejected_enabled	Boolean	No	Indicates whether to "wait" for the image data to be "ejected" by the press/printer. Default value is **false**. Only "enable" if modifying this FITS-Output plug-in to "transfer" image data to print electronics.

Table 3: FITS-Output-Raw Settings

Element	Type	Required	Purpose
output_tiff	JSON Object	No	JSON Object containing the attributes related to the FITS-Output-Tiff plug-in.
directory	String	No	Indicates the directory where image data files are written (if enabled). Default value is "**./raster**".
file_enabled	Boolean	No	Indicates whether the image data is written to file. Default value is **false**.
job_name_enabled	Boolean	No	If writing to file, indicates whether the Fiery job name is used in the name of the output file. Default value is **false**.
colorant_accumulator_enabled	Boolean	No	Indicates whether the image data is "accumulated" and processed in a single thread. Default value is **false**.
wait_for_ejected_enabled	Boolean	No	Indicates whether to "wait" for the image data to be "ejected" by the press/printer. Default value is **false**. Only "enable" if modifying this FITS-Output plug-in to "transfer" image data to print electronics.

Table 4: FITS-Output-Tiff Settings

Element	Type	Required	Purpose
output_prn	JSON Object	No	JSON Object containing the attributes related to the FITS-Output-Prn plug-in.
directory	String	No	Indicates the directory where image data files are written (if enabled). Default value is "**./raster**".
file_enabled	Boolean	No	Indicates whether the image data is written to file. Default value is **false**.
job_name_enabled	Boolean	No	If writing to file, indicates whether the Fiery job name is used in the name of the output file. Default value is **false**.
wait_for_ejected_enabled	Boolean	No	Indicates whether to "wait" for the image data to be "ejected" by the press/printer. Default value is **false**. Only "enable" if modifying this FITS-Output plug-in to "transfer" image data to print electronics.

Table 5: FITS-Output-Prn Settings

Element	Type	Required	Purpose
output_simulator	JSON Object	No	JSON Object containing the attributes related to the FITS-Output-Simulator plug-in.
images_per_minute	Number	No	Indicates the image rate (per minute) to simulate. Default value is **0** (i.e. disabled).
colorant_accumulator_enabled	Boolean	No	Indicates whether the image data is "accumulated" and processed in a single thread. Default value is **false**.
transfer_time_in_milliseconds	Number	No	Indicates the amount of time (in milliseconds) each image should delay to simulate data "transfer" to the electronics of a press/printer. Default value is **0** (i.e. disabled).

Table 67: FITS-Output-Simulator Settings

4 Customization

The sample FDX FITS-Client is designed as a framework that printer clients can leverage to 1) build upon the existing FITS-Output plug-in, 2) build their own FITS-Output plug-in, 3) build their own FITS-Module, or 4) build their own FITS-Client application. Modifying an existing FITS-Output plug-in or building your own is clearly the least involved way to customize FDX FITS-Client to suit your needs. Modifying the existing FITS-Module (including any FITS-Output plug-ins -- if the interface is modified) or building your own FITS-Module is more involved, while modifying the existing FITS-Core (including the FITS-Module and any FITS-Output plug-ins -- if the interfaces are modified) or building your own FITS-Client is the most involved. Carefully weigh your needs to understand which form of customization will work best for you.

4.1 FITS-Output

If writing your own FITS-Output plug-in, there are 5 methods that must be defined. The actual details of the APIs are found in FITS-OutputApi.h, but in general terms, the methods are:

- Initialize: Initializes the plug-in and informs the FITS-Module which of the remaining APIs are “enabled” (and can thus be invoked by the FITS-Module).
- ImageProcess: Allows a FITS-Output plug-in to “process” the given image data before being stored in memory.
- ImageTransfer: Allows a FITS-Output plug-in to “transfer” the given image data for a given colorant.
- ImageTransferAccumulated: Allows a FITS-Output plug-in to “transfer” the given image data for all colorants in a single thread.
- ImageWaitForEjected: Allows a FITS-Output plug-in to “wait” for the given image to be “ejected” by the electronics of a printer – ensuring the image will not need to be recovered/reprinted if a problem arises on the printer.

Note that even though a plug-in may support both ImageTransfer() and ImageTransferAccumulated(), only one of them can be “enabled” at a time.

One reason modifying an existing FITS-Output plug-in (or writing your own) is the least involved way to customize FDX FITS-Client, is that it does not require re-building the FITS-Core or FITS-Module – you can simply use the binaries provided in the DataXPress FITS-Client release/package.

The following sections describe how some of the existing FITS-Output plug-ins implement and use the 5 required methods.

4.1.1 FITS-Output-Raw

FITS-Output-Raw is intended to function without “processing” the image data – thus, ImageProcess() is “disabled”. Both ImageTransfer() and ImageTransferAccumulated() are supported, with ImageTransferAccumulated() calling ImageTransfer() for each colorant’s image data. ImageTransfer() has an “INSERT CODE HERE” comment where code can be added to implement the actual “transfer” of image data to your print electronics. The “colorant_accumulator_enabled” configuration setting controls which “transfer” method is “enabled”. Similarly, ImageWaitForEjected() also has an “INSERT CODE HERE” comment where code can be added that will “wait” for the given page to be “ejected” your print electronics.

4.1.2 FITS-Output-Tiff

FITS-Output-Tiff is intended to “process” the image data into TIFF format prior to being stored in memory – thus, ImageProcess() is “enabled”. Both ImageTransfer() and ImageTransferAccumulated() are supported, with both having an “INSERT CODE HERE” comment where code can be added to implement the actual “transfer” of image data to your print electronics. The “colorant_accumulator_enabled” configuration setting controls which “transfer” method is “enabled”. Similarly, ImageWaitForEjected() also has an “INSERT CODE HERE” comment where code can be added that will “wait” for the given page to be “ejected” your print electronics.

4.1.3 FITS-Output-Prn

FITS-Output-Prn requires an “accumulated” image before “generating” the PRN format – thus, both ImageProcess() and ImageTransfer() are “disabled”. ImageTransferAccumulated() is supported, with ImageTransferAccumulated() first “generating” the PRN format and then attempting to “transfer” the PRN image data. ImageTransferAccumulated() has an “INSERT CODE HERE” comment where code can be added to implement the actual “transfer” of image data to your print electronics. Similarly, ImageWaitForEjected() also has an “INSERT CODE HERE” comment where code can be added that will “wait” for the given page to be “ejected” your print electronics.

4.2 FITS-Module

If replacing the FITS-Module, there are 4 methods that must be implemented. The actual details of the APIs are found in FITS-ModuleApi.h, but in general terms, the methods are:

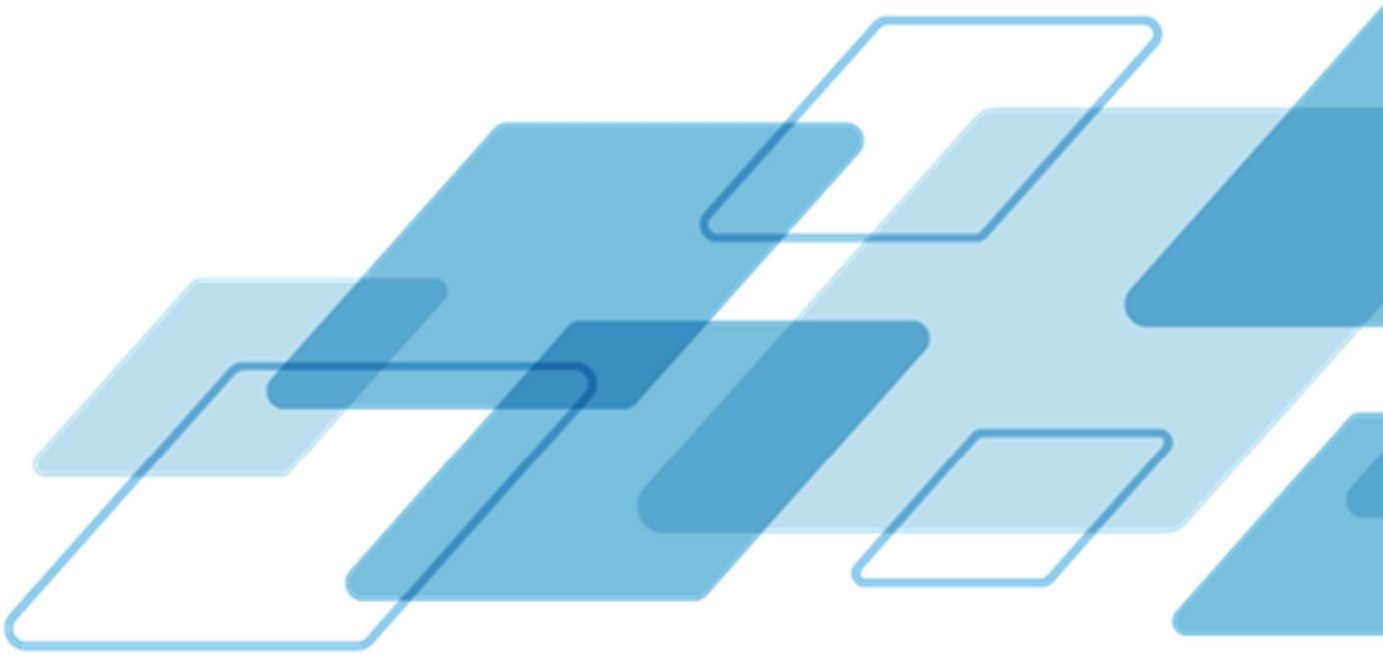
- Initialize: Initializes the FITS-Module.
- Start: Starts the FITS-Module interface.
- Stop: Stops the FITS-Module interface.
- ImageSave: Allows the FITS-Module to “save” the given image data for eventual “transfer”.

Although there are fewer “required” methods than a FITS-Output plug-in, there is a larger amount of internal architecture that must be implemented. This includes an image “buffering” mechanism to “store” images until your FieryPath client requests a “print” (or “cancel”), an image “transfer” mechanism to send image data to your print electronics, and a “communication” mechanism to communicate with your FieryPath client to manage “print” and “cancel” requests.

One reason replacing the FITS-Module is less involved than replacing the entire FITS-Client, is that it does not require re-building the FITS-Core – you can simply use the binary provided in the DataXPress FITS-Client release/package.

4.3 FITS-Core

If replacing the FITS-Core, in addition to implementing the internal architecture required with replacing the FITS-Module, the other requirement is implementing the FITS protocol as described in **FieryPath_FITS** specification. Thus, making this the most involved way to customize the FDX FITS-Client.



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